MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5670W 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, miniature, medium Mu

Outline	---	6-1 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	h	2k	2g	2a	sd	1a	1g	1k	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk/k	Rg/g	Ik/k	Ic/g	Pp/p	TE	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	mAdc	W	°C	ft
Maximum	6.9	330	0,-55	100	---	0.5	18	3.0	1.35	+165	(See note 1)
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	150	0	0	240	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

1/ Formerly tube type 5670

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METHOD	REQUIREMENTS OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.4	II	If	330	370	mA
1336	Heater-cathode leakage	See note 3	0.4	II	Ihk	---	10	µAdc
1266	Total grid current	Rg = 0.5 Meg (see notes 2 and 3)	0.4	II	Ic	0	-0.3	µAdc
1256	Electrode current (1) (anode)	See note 3	0.4	II	Ib	5.9	10.5	mAdc
1256	Electrode current (2) (anode)	Ec =-10 Vdc; Rp = 0.25 Meg (see note 3)	0.4	II	Ib	---	45	µAdc
1306	Transconductance (1)	See note 3	0.4	II	Sm	4,500	6,500	µmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	See note 3	2.5	I	---	---	---	---
1256	Electrode current (1) (anode) (difference between sections)		2.5	I	Ib	---	1.8	mAdc
1256	Electrode current (3) (anode)	Ec = -4 Vdc (see note 3)	2.5	I	Ib	5	---	µAdc
1306	Transconductance (2)	Ef = 5.7 V (see note 3)	2.5	I	ΔSm Ef	---	15	%
1266	Grid emission	Ef = 7.5 V; Rg = 0.5 Meg; Ec = -10 Vdc (see notes 3 and 4)	2.5	I	Isc	0	-0.5	µAdc
2201	Noise and microphonics	Ef = 6.3 Vac; Ehk = 0; Ebb= 250 Vdc; Ec= 0; Ecal = 200 mVac; Rp = 10,000 ohms; Rg=0; Rk = 240 ohms (see note 7)	2.5	I	---	---	---	---
1316	Amplification factor	See note 3	6.5	S3	Mu	26	44	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
1331	Direct-inter-electrode capaci-tance	No shield (note 3)	6.5	Code E	Cgp	0.8	1.4	Pf
		No shield (note 3)			Cin	1.7	2.7	pF
		No shield (note 3)			Cout	0.7	1.3	pF
		No shield			Cpp	---	0.10	pF
1031	Low-frequency vibration	Ec= -3 Vdc; Rp= 2,000 ohms; 10 G; F = 40 Hz; Rk = 0 (see note 7)	6.5	Code H	Ep	---	130	mVac
---	Pulse cathode current	Ecl = -20 Vdc; Rk = 1.0 ohm (see notes 3 and 10)	2.5	Code H	ik	270	---	ma
1041	Shock	630 G; Ehk = +100 Vdc; Ck = 0 (see note 5)	---	See note 11	---	---	---	---
1031	Vibration fatigue	2.5 G; fixed frequency; F = 25 (min), 60 (max)	6.5	See note 11	---	---	---	---
---	Post-shock and vibration-fatigue test end points	Low-frequency vibration	---	---	Ep	---	200	mVac
		Heater-cathode leakage	---	---	Ihk	---	20	µAdc
		Transconduct-ance (1)	---	---	Sm(1)	3,850	---	µmhos
		Total grid current	---	---	Ic	0	-0.6	µAdc
1121	Base strain	See note 8	---	---	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
1506	Heater cycling life	Ef = 7.5 V Ehk = +135 Vdc; Rk = 0; Ec = Eb = 0	---	---	---	---
---	Heater-cycling life test end point		---	---	---	---
1336	Heater-cathode leakage		Ihk	---	15	µAdc
1516	Stability life	Ehk = +135 Vdc; Rg/g = 0.5 Meg; TA = room	---	---	---	---
---	Stability life-test end point		---	---	---	---
1306	Change in trans-conductance (1) of individual tubes		ΔSm t	---	10	%
1501	Intermittent life	Stability life-test conditions; TE = +165°C (min) (see notes 6 and 9)	---	---	---	---
---	Intermittent life-test end points (1,000 hours)		---	---	---	---
1266	Inoperatives		---	---	---	---
1266	Total grid current		Ic	0	-0.3	µAdc
1301	Heater current		If	330	380	mA
1306	Change in transconduct-ance (1) of individual tubes		ΔSm t	---	25	%
1306	Transconductance (2)		ΔSm Ef	---	20	%
1336	Heater-cathode leakage		Ink	---	10	µAdc
1211	Insulation of electrodes		R	50	---	Meg

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. This test shall be performed at the conclusion of the holding period.
3. Test each unit separately.
4. Prior to this test, tubes shall be preheated a minimum of 5 minutes with each section operating at the conditions specified below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef V	Ec Vdc	Eb Vdc	Rk/k Ohms	Rg/g Meg
7.5	0	-150	240	0.5

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NOTES: -Continued.

5. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thratron-type short indicator is employed.

6. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

7. Tie 1k to 2k; 1g to 2g; and 1a to 2a.

8. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

   (a) Accepted if not more than one defective for class "A", "B", or "C" defects, respectively (see method 1121), or if not more than a total of two defectives are found in the sample.

   (b) Rejected if two or more defectives for class "A", "B", or "C" defects, respectively, or if a total of three or more defectives are found in the sample.

9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

10. The grid is driven with a pulse voltage as follows: egk = +30 v; prr = 1,000; tp = 10 µs; tr < 1 µs; tf < 1 µs. Peak cathode current shall be measured by means of a high impedance oscilloscope, or equivalent device, connected across a cathode resistor of 1.0 ohm. Preheat at Ef = 6.3 V for 5 minutes; no other voltages applied.

11. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 12-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.

12. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
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Navy - EC
Air Force - 85

Review activities:
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Air Force - 99
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Preparing activity:
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